higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 4

(6021034)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only a BLACK or BLUE pen.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 An air-conditioning condenser is used to….

A cool and remove the heat from the refrigerants
B operate as a blower, motor and fan installed inside the vehicle
C increase the pressure and temperature at the same time
D purify the air in the vehicle's interior by removing the dust particles

1.2 A compressor is used to …

A cool and remove the heat from the refrigerants
B operate as a blower, motor and fan installed inside the vehicle
C increase the pressure and temperature at the same time
D purify the air in the vehicle's interior by removing the dust particles

1.3 The air purifier is used to …

A cool and remove the heat from the refrigerants
B operate as a blower, motor and fan installed inside the vehicle
C increase the pressure and temperature at the same time
D purify the air in the vehicle's interior by removing the dust particles

1.4 Port fuel injection occurs when fuel …

A is injected into the combustion chamber.
B is injected into the intake manifold.
C enters the airstream in the air horn.
D is sprayed towards the intake valve.

1.5 To rectify excessive heel contact, you need to adjust the …

A pinion towards the centre-line of the crown wheel.
B crown wheel away from the pinion gear.
C pinion away from the centre-line of the crown wheel.
D crown wheel towards the pinion gear.

1.6 To rectify excessive toe contact, you need to adjust the …

A pinion towards the centre-line of the crown wheel.
B crown wheel away from the pinion gear.
C pinion away from the centre-line of the crown wheel.
D crown wheel towards the pinion gear.

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1.7 To rectify excessive flank contact, you need to adjust the …

A pinion towards the centre-line of the crown wheel.
B crown wheel away from the pinion gear.
C pinion away from the centre-line of the crown wheel.
D crown wheel towards the pinion gear.

1.8 To rectify excessive face contact, you need to adjust the …

A pinion towards the centre-line of the crown wheel.
B crown wheel away from the pinion gear.
C pinion away from the centre-line of the crown wheel.
D crown wheel towards the pinion gear.

1.9 To measure the axial run-out, you must fit the dial indicator needle …

A to the circumference of the crown wheel.
B on the cross shaft.
C on the side of the crown wheel.
D between the sun gear and planetary gear.

1.10 To measure the radial run-out, you must fit the dial indicator needle…

A to the circumference of the crown wheel.
B on the cross shaft.
C on the side of the crown wheel.
D between the sun gear and planetary gear.

(10 × 1) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 The main journal turns twice the speed of the big-end journal of the crank shaft.

2.2 Too much torque will overcome the friction and the vehicle's wheels will spin.

2.3 Auxiliary equipment is powered by an HV-battery.

2.4 The coefficient of friction is determined by the stickiness between the two surfaces.

2.5 A hybrid electrical car uses 6 V and 12 V conventional batteries.

(5 × 1) **[5]**

QUESTION 3

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–M) next to the question number (3.1–3.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	Single-speed double reduction	A	allows for expansion between the crown wheel and pinion
3.2	Double-speed double reduction	B	lowers the refrigerant pressure
3.3	Backlash	C	reduces fluorine-gas emissions
3.4	MAC-directive	D	has two pinion gears
3.5	Stator	E	refers to ozone-depleting substances
3.6	Torque convertor	F	consists of an conventional crown wheel and pinion
3.7	F-gas	G	refers to hydrochlorofluorocarbons
3.8	Expansion valve	H	aims to reduce emissions from certain types of gasses
3.9	ODS	I	changes the direction of the oil leaving the turbine
3.10	HCFC	J	chlorofluorocarbons
		K	converts potential energy into kinetic energy
		L	performs the same basic function as a clutch in the manual transmission
		M	refers to mobile air-conditioning system

(10 × 1) **[10]**

QUESTION 4

Complete the following paragraph by using the words in the list below. Write only the missing word or words next to the question number (4.1–4.5) in the ANSWER BOOK.

planetary; distance; cross shaft; slower; time; sun gear; faster; speed

When a vehicle makes a right turn, the inner wheel turns (4.1) … than the opposite wheel. Both wheels must turn around the corner at the same (4.2) …. The outer wheel must turn a greater (4.3) … than the opposite wheel. To make the difference in speed possible, the (4.4) … gears are forced to rotate around the (4.5) … (5 × 1) **[5]**

TOTAL SECTION A: 30

SECTION B

QUESTION 5

5.1 Explain EACH of the following principles of lubrication related to the differential during operation:

5.1.1 Method of lubrication

5.1.2 Function of lubrication

5.1.3 Type of lubrication

(3 × 1) (3)

5.2 Explain the operation of a single-speed double-reduction differential. (3)

5.3 Explain FIVE safety practices and procedures when working with air-conditioning systems. (5)

5.4 Differentiate between the types of devices that are used in air-conditioning systems to lower refrigerant pressure. (2)

5.5 Explain how to perform a stall test to identify possible mechanical or hydraulic faults on an automatic transmission. (4)

5.6 Give THREE advantages of the automatic transmission. (3)

[20]

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QUESTION 6

6.1 Testing and adjusting the electronic ignition system (EIS) is very similar to testing and adjusting the electronic fuel injection (EFI) system. Most of the tests and procedures are the same. This is because the EFI-system works in conjunction with the EIS. The one system is dependent on the other to operate effectively. It is very important to wear PPE at all times when working on the EIS and EFI.

6.1.1 Differentiate between an *electronic ignition system* and an *electronic fuel injection* system. (2)

6.1.2 State FOUR safety steps, in the correct sequence, that a technician must follow when working on an electronic ignition system.
The answer must exclude PPE, but include the following: ECU harness, digital meter, etc. (4)

6.1.3 Explain the FOUR steps to check for fuel blockages in the line without blowing air in the lines. (4)

6.2 Differentiate between the following in a diesel fuel system:

6.2.1 Operating pressure and retention pressure (2 × 2) (4)

6.2.2 Mechanical electronic injector and hydraulic energy injector (2 × 2) (4)

6.3 Explain how to test the fuel flow rate on a diesel fuel system. (3)

6.4 Explain the FOUR steps to follow when testing an electronic fuel system for air and fuel leakages. (4)

[25]

QUESTION 7

7.1 Name any THREE parts of a rack-and-pinon steering system. (3)

7.2 Differentiate between an *electric hydraulic steering pump* and an *engine-driven steering pump.* (2)

7.3 Give THREE advantages of an electric-powered steering pump. (3)

7.4 List the functions of EACH of the following advanced diagnostic tools used to trace faults:

7.4.1 Scanning tool

7.4.2 Lab scope

(2 × 2) (4)

7.5 Differentiate between the following:

7.5.1 Kinetic friction and static friction on brake systems

7.5.2 Braking force and accelerating force

7.5.3 Traction-control system (TCS) and electronic stability control (ESC)
(3 × 2) (6)

7.6 Give TWO advantages of a hybrid system. (2)

7.7 The complete supplementary restrain system (SRS) consists of different parts that work together to contribute to the safety of the occupants in a vehicle.

List FOUR parts of the SRS system. (4)

7.8 Indicate whether the following statement is TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (7.8) in the ANSWER BOOK.

If one part of the SRS fails, the complete system will be deactivated. (1)
[25]

TOTAL SECTION B: 70
GRAND TOTAL: 100

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